CURIOUS SPAROOWS · ENCYCLOPEDIC INDEX · NEEDLE ARTS

SEWING

How a needle with a hole in it changed human history

Machines · tools, threads and fabrics · techniques · pattern making and fit · 24 garments register · quilting, mending and more · history · designers and awards · ideas · practical guide

Parts	Sections	Sub-sections	Entries	Register rows
13	27	5	133	47

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How to Use This Booklet

This booklet is a map of sewing: machines, tools and materials, techniques, pattern making, garments, quilting and mending, history, people, ideas and a practical guide.

Every entry has an ID (such as SEW-07.1 for darts). The same ID is the entry's address on curioussparoows.com.

How this booklet is organised

Parts 0 to 12: intro questions, map, machines, tools and materials, techniques, pattern making, registers, beyond garments, history and people, ideas, practical guide, reviews, navigation and sources.

Entries (☐): one or two lines each; tick when learned or made.

Registers: 24 garments and the timeline.

How to explore it

Step 1. Read Part 0.

Step 2. Learn your machine and its parts (Part 2).

Step 3. Make the first 10 projects in order (Part 10).

Step 4. Pick one garment from the register and fit it to yourself.

Step 5. Tick ☐ and keep a sewing log.

Our promise

Knowledge belongs to the people who created it. Patterns are described, never copied; designers, tailors and karigars are credited. Where something could not be verified, we say so instead of guessing.

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PART 0 — THE INTRODUCTION TWIG: HOW DID A NEEDLE WITH A HOLE IN IT CHANGE HUMAN HISTORY?

About this Part. The doorway to the Sewing section of the website, in the **Needle Arts** branch with Crochet (CRO), Knitting (KNT) and Embroidery (EMB). One Gateway page, five Leaves and one Recap; each Leaf takes about two minutes.

SEW-00 The Gateway: How did a needle with a hole in it change human history?

Web address: /sewing/needle-with-a-hole/ · **Keywords:** sewing for beginners, history of sewing, how a sewing machine works, Indian tailoring, slow fashion

A sharp stone can make a hole in a hide. A needle with an eye does something more: it carries the thread through by itself. With that small idea, people could join skins tightly, make fitted clothes, survive ice ages, and later decorate what they wore. Thousands of years later, the same idea, moved to the tip of a machine needle, built the ready-made garment industry that dresses most of the world, including the tailors' shops on every Indian street.

This short course asks one question in five steps: **how did sewing shape the way we live?**

By the end you will be able to:

- Say how old eyed needles are and why they mattered.
- Explain why so much traditional Indian dress is unstitched, and where stitched clothes came from.
- Describe how a sewing machine ties a lockstitch with two threads.
- Explain why the sewing machine was called a "freedom machine", and where that story is too simple.
- Argue for and against the idea that sewing can fight fast fashion.

SEW-00.01 Who made the first needle?

> **Short answer:** Nobody knows the person, but we know the age. Bone needles with an eye appear at least about 40,000 years ago; one from Denisova Cave in Siberia is often reported at around 50,000 years, though that date is debated. Earlier people used pointed awls to punch holes and pushed sinew through by hand.

Keywords: first needle, oldest bone needle, Denisova needle, prehistoric sewing, eyed needle

What the needles tell us

- **Awls first, eyes later**: an awl makes the hole; an eyed needle makes the hole and pulls the thread in one movement, so seams become finer and tighter.
- **Warm, fitted clothes**: tight seams kept out wind and cold, which helped people live in the far north during the ice ages.
- **More than survival**: a 2024 study in *Science Advances* (Gilligan and colleagues) argued that very fine eyed needles also point to decorated clothing, that is, dress as a sign of who you are.
- **Thread**: sinew, plant fibres and later spun yarn. Spinning and sewing grew up together.
- **India**: copper and bronze needles are found at Harappan sites, and later texts and sculptures show both stitched and draped dress.

Remember this: The eye of the needle is one of humanity's great small inventions: it turned "a hole and a string" into a seam.

Next: SEW-00.02 Why did India love unstitched clothes?

SEW-00.02 Why did India love unstitched clothes?

Short answer: In a hot climate, a long uncut cloth such as the sari, dhoti or lungi is cool, fits any body and any age, and can be re-wrapped many ways. In many communities an uncut, unstitched cloth was also felt to be pure and right for worship. Stitched clothes were always present too, and grew with contact with Central Asia, Persia and later Europe.

Keywords: unstitched clothing India, sari history, dhoti, stitched vs draped, history of Indian clothing

The two traditions

- **Draped (unstitched)**: sari, dhoti, lungi, mundu, uttariya (upper cloth), turban (pagri, safa). One loom, one length, endless styles of wrapping.
- **Stitched**: tunics appear in Kushan-period sculpture; the jama, angarkha and pyjama spread under the Sultanate and Mughal courts; the sari blouse and petticoat became common in the 19th century (the blouse style is often credited to Jnanadanandini Devi in 1870s Bengal).
- **Purity and ritual**: many temples and rituals still ask for unstitched cloth, such as a dhoti for priests or the two white sheets of the Hajj *ihram*.
- **Today**: most Indians wear both, a stitched kurta or shirt with a draped dupatta or sari, and the local darzi (tailor) stitches the blouse to measure.

Remember this: India was not "behind" in sewing; it chose draping where draping worked better, and stitching where stitching did.

Next: SEW-00.03 How does a sewing machine tie a knot with two threads?

SEW-00.03 How does a sewing machine tie a knot with two threads?

Short answer: The machine needle has its eye near the point. It pushes the top thread down through the cloth and makes a small loop underneath. A rotating hook (or a shuttle in old machines) catches that loop and passes it round the bobbin, which holds the second thread. The take-up lever then pulls the loop tight, so the two threads lock together inside the fabric. This is the **lockstitch**.

Keywords: how a sewing machine works, lockstitch, bobbin, rotary hook, chain stitch machine

The parts that work together

- **Needle**: eye at the point, a groove for the thread, a flat side so it sits the right way.
- **Hook and bobbin**: the hook grabs the loop; the bobbin feeds the lower thread.
- **Take-up lever**: moves up and down to pull the stitch tight and draw new thread.
- **Feed dogs**: small teeth under the presser foot that move the cloth forward one stitch at a time.
- **Tension discs**: balance the two threads so the "knot" sits in the middle of the cloth, not on top or below.
- **Chain stitch, the other way**: the first machines and today's sacks and overlockers use loops of thread linked like a crochet chain, which can unravel if pulled from one end.

Remember this: A hand needle goes all the way through; a machine needle never does. The machine's genius is the needle with an eye at its point and the hook that catches its loop.

Next: SEW-00.04 Why was the sewing machine called a freedom machine?

SEW-00.04 Why was the sewing machine called a freedom machine?

Short answer: In the late 1800s the home sewing machine cut the hours of hand-stitching that fell mainly on women, and Singer's instalment plan let ordinary families buy one. Machines also let women earn money from home. But the same machine made factory sweatshops possible. It freed some women and tied others to long hours for low pay.

Keywords: history of the sewing machine, Singer, Elias Howe, women and sewing machine, sweatshop history

Both sides of the story

- **Time saved**: a shirt that took many hours by hand could be stitched in about an hour by machine.
- **Credit**: Singer and his partner Edward Clark sold machines on hire purchase from the 1850s, one of the first big consumer-credit schemes.
- **Livelihood**: tailoring and stitching became home income for millions, including women in India today, where the sewing machine is a common gift in skill and welfare schemes.
- **The dark side**: garment factories with long hours, from New York's Triangle Shirtwaist Factory fire (1911) to the Rana Plaza collapse in Bangladesh (2013), where more than 1,100 garment workers died.
- **Gandhi's view**: he is reported to have praised the Singer machine as one of the few useful things ever invented, because it served the person working it.

Remember this: A tool is "freedom" only when the person using it controls the time and the money it makes.

Next: SEW-00.05 Can sewing fight fast fashion?

SEW-00.05 Can sewing fight fast fashion?

Short answer: Partly. Fast fashion makes clothes cheap and short-lived; the Ellen MacArthur Foundation (2017) found that clothing production roughly doubled between 2000 and 2014, and that less than 1% of the material used for clothing is recycled into new clothing. Sewing helps by making clothes last longer through mending, altering, upcycling and buying fewer, better-made pieces. It cannot fix the system alone; brands and governments must change too.

Keywords: fast fashion, slow fashion, textile waste, visible mending, upcycling clothes

What a needle can do

- **Mend**: a button, a seam, a patch. India's culture of repair (the darzi, the *rafoo* darner, the patched godhadi quilt) is a model.
- **Alter**: take in, let out, shorten; Indian tailors leave wide seam allowances for this.
- **Upcycle**: old saris into quilts, kurtas into cushion covers, jeans into bags.
- **Make slowly**: one garment that fits, in natural fabric, worn for years.
- **The limits**: home sewing is small next to global production; repair also needs time, skill and cheap thread, which many do not have.

Remember this: The greenest garment is the one you already own; the needle keeps it alive.

Next: SEW-00.06 Recap.

SEW-00.06 Recap: the five steps in one page

- **Needle (SEW-00.01)**: eyed needles at least about 40,000 years old turned holes into seams.
- **India (SEW-00.02)**: draped and stitched dress grew side by side for good reasons.
- **Machine (SEW-00.03)**: a needle with an eye at its point plus a hook makes the lockstitch.
- **Freedom (SEW-00.04)**: the machine saved time and made livelihoods, and also made sweatshops.
- **Future (SEW-00.05)**: mending, altering and slow making help against fast fashion.

Try this: Find one garment at home with a loose button or open seam and fix it by hand this week (SEW-11.5 and SEW-15).

PART 1 — THE MASTER MAP OF SEWING

SEW-01 The hierarchy: from Forest to Leaf

Before you start — SEW-01. The Forest is "Sewing". Its Trees are machines, tools and materials, techniques, pattern making and fit, garments, other kinds of sewing, and the human side (history, people, ideas, practice). **See also:** CRO-01, KNT-01, EMB-01.

☐ **SEW-01.1 Machines** (Part 2).
☐ **SEW-01.2 Tools and materials**: needles, threads, fabrics, interfacings, notions (Part 3).
☐ **SEW-01.3 Techniques**: stitches, seams, shaping, openings, finishing, pressing (Part 4).
☐ **SEW-01.4 Pattern making and fit** (Part 5).
☐ **SEW-01.5 Registers**: garments and the timeline (Part 6).
☐ **SEW-01.6 Beyond garments**: home, bags, quilts, mending, costume (Part 7).
☐ **SEW-01.7 The human side**: history, people and awards (Part 8), ideas (Part 9), practical guide (Part 10), reviews (Part 11).

PART 2 — SEWING MACHINES

SEW-02 Kinds of machine

☐ **SEW-02.1 Hand-crank machine**: turned by a handle; cheap, needs no power, still common in Indian homes and villages.

☐ **SEW-02.2 Treadle machine**: worked by the feet on an iron stand, leaving both hands free; the classic black Usha or Singer on a wooden table, used by many darzis.

☐ **SEW-02.3 Electric (mechanical) machine**: a motor and foot pedal; stitches chosen by dials. Most home learners start here.

☐ **SEW-02.4 Computerised machine**: stitches chosen on a screen, automatic buttonholes, needle threader, speed control; more features, more to repair.

☐ **SEW-02.5 Overlocker (serger)**: trims and wraps the raw edge with 3 to 5 threads in one pass; the neat inside seam of ready-made T-shirts.

☐ **SEW-02.6 Coverstitch machine**: the double-row hem on knit garments.

☐ **SEW-02.7 Industrial machines**: single-needle lockstitch, overlock, flatlock, buttonhole and bar-tack machines that run all day at high speed; each does one job very well.

☐ **SEW-02.8 Embroidery and quilting machines**: computerised embroidery units (EMB-02.8) and long-arm quilting machines on frames.

☐ **SEW-02.9 The Indian story: Singer and Usha**: Singer machines reached India in the late 19th century; Usha machines were made in Kolkata from 1935, and Indian brands (Usha, later Jack, Brother and others in industry) put a machine in millions of homes. In India the machine is also a dowry item, a welfare gift and a small business.

SEW-03 Parts, care and troubleshooting

☐ **SEW-03.1 Main parts**: spool pin, thread guides, tension discs, take-up lever, needle bar and clamp, presser foot and lifter, feed dogs, throat (needle) plate, bobbin, bobbin case and hook, hand wheel, stitch length and width controls, reverse lever.

☐ **SEW-03.2 Presser feet**: all-purpose, zipper foot, buttonhole foot, blind-hem foot, walking (even-feed) foot, rolled-hem foot, piping foot.

☐ **SEW-03.3 Care**: brush out lint after each project; oil where the manual says (old machines need more); change the needle every 8 or so hours of sewing; cover the machine against dust.

☐ **SEW-03.4 Troubleshooting**: thread nests underneath → rethread the top with the foot **up**; skipped stitches → new or right-type needle; puckering → shorter stitch or less tension; needle breaking → bent needle or pulling the cloth; loops on top → bobbin tension or wrong bobbin.

PART 3 — TOOLS AND MATERIALS

SEW-04 Tools

☐ **SEW-04.1 Hand needles**: sharps (general), betweens (quilting), crewel (embroidery), ballpoint, darners; sizes get finer as the number rises.

☐ **SEW-04.2 Machine needles**: the home system is 130/705H; sizes 70/10 (fine) to 110/18 (heavy). Types: universal, ballpoint or jersey (knits), stretch, microtex or sharp (silk, tight weaves), denim, leather, twin, topstitch.

☐ **SEW-04.3 Cutting**: dressmaking shears (only for cloth), paper scissors, snips, pinking shears, rotary cutter and mat, seam ripper (*the* most used tool).

☐ **SEW-04.4 Measuring and marking**: inch tape, ruler, French and hip curves, tailor's chalk, washable pens, tracing wheel and carbon.

☐ **SEW-04.5 Pressing**: iron with steam, pressing cloth, tailor's ham, sleeve roll, point presser, clapper.

☐ **SEW-04.6 Others**: pins and pin cushion, thimble, bodkin, loop turner, dress form, pattern paper.

SEW-05 Threads, fabrics and notions

☐ **SEW-05.1 Threads**: polyester all-purpose (strong, slight stretch), cotton (for cotton, does not melt under the iron), silk (fine work, tailoring), topstitch, elastic shirring, overlocker cones; thickness given as Tex or weight.

☐ **SEW-05.2 Woven fabrics**: threads cross at right angles; grain (lengthwise), crossgrain and bias (45°, stretchy and drapey); poplin, cambric, lawn, voile, denim, twill, canvas, linen.

☐ **SEW-05.3 Knit fabrics**: looped (KNT-01), stretchy; jersey, rib, interlock; need ballpoint needles and stretch stitches.

☐ **SEW-05.4 Indian fabrics**: **khadi** (hand-spun and hand-woven, Gandhi's cloth of self-reliance); **mul** or mulmul (soft fine cotton); **cambric** and **poplin** (kurtas and shirts); **rubia** (lining voile); **casement** and **2x2 rubia** for blouses; **silks** (Banarasi, Kanchipuram, tussar, mulberry, eri, muga); **chanderi** and **maheshwari** (silk-cotton); **ikat** and **ajrakh**; **georgette** and **chiffon** (often synthetic now).

☐ **SEW-05.5 Preparing cloth**: pre-wash to shrink, straighten the grain, iron, then cut. Most "my kurta shrank" stories start here.

☐ **SEW-05.6 Interfacings**: fusible (iron-on) or sew-in layers that stiffen collars, cuffs, plackets and waistbands; Indian tailors call buckram and canvas *kaccha* and *pakka*.

☐ **SEW-05.7 Notions**: buttons, hooks and eyes (the blouse *hook*), press studs, zips (invisible, metal, nylon; Gideon Sundback's "hookless fastener" was patented in 1917), elastic, drawstrings (*nada*), piping, bias tape, lace, latkan (tassels).

PART 4 — TECHNIQUES

SEW-06 Stitches and seams

☐ **SEW-06.1 Hand stitches**: running, tacking (basting, *kacchi silai*), back stitch, slip stitch, hemming stitch, whip stitch, blanket stitch, catch (herringbone) stitch, buttonhole stitch.

☐ **SEW-06.2 Machine stitches**: straight (length 2.5 mm is a good start), zigzag, stretch (lightning), blind hem, overlock-style, buttonhole.

☐ **SEW-06.3 Seam allowance**: commercial patterns use 1.5 cm (5/8 inch); Indian tailors often leave 2.5 cm or more in kurtas and blouses for later alteration.

☐ **SEW-06.4 Seam types**: plain seam, French seam (raw edge hidden inside, for sheer fabric), flat-felled seam (jeans, shirts), bound or Hong Kong seam, lapped seam.

☐ **SEW-06.5 Seam finishes**: pinking, zigzag, overlocking, turned-and-stitched, binding. The inside of a garment shows its quality.

SEW-07 Shaping, openings and finishing

☐ **SEW-07.1 Darts**: folded wedges that turn flat cloth into 3D shape at bust, waist and back; the heart of the blouse.

☐ **SEW-07.2 Pleats, tucks and gathers**: knife, box and inverted pleats; pin tucks; gathers for skirts, lehengas and puffed sleeves; *kalis* (panels) in anarkalis and lehengas.

☐ **SEW-07.3 Hems**: double-fold, blind hem, rolled hem (dupattas, sarees), faced hem, *fall* stitching on a sari.

☐ **SEW-07.4 Facings and binding**: finishing necklines and armholes; Indian kurtas use shaped facings for "U", "V", boat and *patti* (placket) necks.

☐ **SEW-07.5 Openings and fastenings**: plackets, zips (centred, lapped, invisible), buttonholes and buttons, hooks, loops (*tukmas*).

☐ **SEW-07.6 Pockets**: patch, side-seam (in kurtas and pyjamas), welt, jeans pockets.

☐ **SEW-07.7 Collars and cuffs**: shirt collar and stand, Chinese or mandarin (*bandhgala*) collar, Peter Pan, shawl collar; cuffs and plackets on sleeves.

☐ **SEW-07.8 Sleeves**: set-in, raglan, kimono, cap, puff, bell; easing a sleeve cap is a classic skill.

☐ **SEW-07.9 Linings**: hide seams, add body and comfort; blouses, lehengas, jackets and coats.

☐ **SEW-07.10 Pressing**: press each seam as you go; pressing is lifting and setting the iron, ironing is sliding it. Tailors say a garment is half stitched and half pressed.

PART 5 — PATTERN MAKING AND FIT

SEW-08 Measure, draft, fit

☐ **SEW-08.1 Measurements**: bust or chest, waist, hip, shoulder, armhole, sleeve length and round, neck depth (front and back), garment length; take them over light clothing, tape snug not tight.

☐ **SEW-08.2 Indian tailoring systems**: the darzi drafts directly on the cloth with chalk from measurements, often in inches, with rules of thumb (such as armhole from chest); taught in ITIs, polytechnic courses and many Hindi and regional "cutting and tailoring" books.

☐ **SEW-08.3 Western flat pattern making**: a basic block (sloper) for bodice, skirt, trousers and sleeve, then altered into styles by moving darts, slashing and spreading. Ebenezer Butterick sold graded tissue-paper patterns from 1863.

☐ **SEW-08.4 Draping**: pinning cloth on a dress form to find the shape; the way couturiers such as Madeleine Vionnet (bias cut) worked.

☐ **SEW-08.5 Grading**: making a pattern in a range of sizes; the basis of ready-made sizing (and why "M" differs between brands).

☐ **SEW-08.6 Fitting**: a toile or muslin trial; common changes for full bust, sway back, sloping shoulders.

☐ **SEW-08.7 Alterations**: shorten hems and sleeves, take in or let out side seams, change buttons, re-set zips; the everyday work of most Indian tailors.

☐ **SEW-08.8 Digital tools**: pattern-making software (CAD), 3D garment simulation, made-to-measure from body scans; used in industry and taught at NIFT.

PART 6 — REGISTERS

SEW-09 Garments register (24)

Before you start — SEW-09. Level: B (beginner), I (intermediate), A (advanced). Time is for a confident home sewer and is only a rough guide. Patterns are described, not copied.

Garment	Level	Fabric	Pattern pieces	Key techniques	Time
Pyjama (drawstring)	B	Cotton, poplin	Two legs (front and back joined), waistband casing	Seams, casing, hem	2 hours
Petticoat (saya)	B	Cotton, poplin	6 to 8 kalis, waistband	Panels, nada casing, hem	2 to 3 hours
Straight kurta	B–I	Cambric, cotton	Front, back, sleeves, facing	Neck facing, side slits, set-in sleeve	4 hours
Salwar (Punjabi)	I	Cotton, rubia	Wide body, side panels (kalis), ankle cuffs	Gathers or pleats, casing	3 hours
Churidar	I	Cotton cut on the bias	Two legs on the bias, belt	Bias cutting, tight fit, ankle buttons	3 hours
Patiala salwar	I	Cotton, crepe	Very wide body, many pleats	Pleating, belt	3 to 4 hours
Anarkali	I–A	Georgette, cotton	Fitted yoke, many kalis	Panels, gathering, lining	6 to 8 hours
Sari blouse	I–A	Cotton, silk with lining	Front (often katori cups), back, sleeves	Darts, fitting, hooks, lining	4 to 6 hours
Lehenga	A	Silk, velvet, net	Many kalis or circle, can-can, waistband	Panels, lining, zip, heavy hem	1 to 2 days
Sherwani	A	Brocade, silk, wool	Long front panels, back, sleeves, collar	Tailoring, interfacing, lining, buttons	2 to 3 days
Nehru (bandhgala) jacket	A	Khadi, wool, linen	Fronts, back, collar stand, lining	Mandarin collar, pockets, lining	1 to 2 days
Angarkha	I–A	Cotton, mul	Overlapping front panels, flared skirt	Ties, curved seams	6 hours
Dhoti pants and modern fusion	B–I	Cotton, rayon	Wide legs with drape, waistband	Pleats, elastic or nada	3 hours
Kids' frock	B	Cotton	Bodice, gathered skirt	Gathers, buttons	3 hours
Tote bag	B	Canvas, denim	Two panels, straps	Straight seams, boxed corners	1 hour
Shirt	I–A	Poplin, cotton, linen	Fronts, back, yoke, sleeves, collar, cuffs	Collar, cuffs, plackets, buttonholes	1 day
Trousers	A	Twill, wool, linen	Fronts, backs, waistband, pockets, fly	Fly zip, pockets, fitting	1 day
A-line skirt	B	Cotton, denim	Front, back, waistband	Darts, zip, hem	3 hours
Circle skirt	B	Cotton, rayon	One or two circles, waistband	Curved hem, bias handling	3 hours
Wrap dress	I	Jersey, rayon	Bodice, skirt, ties	Knits, neck binding	5 hours
Shirt dress	I–A	Cotton, linen	Shirt pieces plus skirt	Collar, buttons, waist	1 day
Jeans	A	Denim	Fronts, backs, yoke, pockets, waistband	Flat-felled seams, rivets, topstitching	1 to 2 days
Unlined jacket	A	Cotton, linen	Fronts, back, sleeves, collar, facings	Set-in sleeves, collar, bound seams	1 to 2 days

Tailored coat	A	Wool	Many pieces plus canvas and lining	Pad stitching, tailoring, lining	Several days

SEW-10 The sewing timeline (year-wise register)

Year	Event	Where	Why it matters
c. 50,000 to 40,000 years ago	Eyed bone needles (Denisova date debated)	Siberia, Eurasia	Fitted clothing
c. 2500 BCE	Copper and bronze needles at Harappan sites	South Asia	Early needlework in India
1st to 3rd centuries CE	Tunics and trousers in Kushan art	North India, Central Asia	Stitched dress in India
16th to 17th centuries	Mughal jama, angarkha, pyjama; court tailors	North India	Stitched court dress
1755	Charles Wiesenthal's British patent for a needle for mechanical sewing	England	First machine-sewing patent
1830	Barthélemy Thimonnier's chain-stitch machine; tailors wreck his workshop in 1831	France	First working machine, first protest
1846	Elias Howe's lockstitch patent	USA	Two-thread lockstitch
1851	Isaac Singer's machine patent	USA	Practical home and factory machine
1856	The Sewing Machine Combination patent pool; Singer's instalment plan	USA	Machines become affordable
1858	Charles Frederick Worth opens his house	Paris	Birth of haute couture
1863	Butterick's graded paper patterns	USA	Home sewing to standard sizes
1889	Singer's electric machine	USA	Motor-driven sewing
1911	Triangle Shirtwaist Factory fire	New York	Labour laws for garment workers
1935	Usha sewing machines in Kolkata	India	Indian-made machines
1947	Dior's "New Look"	Paris	Post-war couture
1975	Singer Athena 2000, first electronic home machine	USA	Computerised sewing
1986	NIFT founded	New Delhi	Fashion education in India
1998	Fashion Design Council of India	New Delhi	Indian fashion weeks
2013	Rana Plaza collapse	Bangladesh	Fashion Revolution movement
2017	*A New Textiles Economy* report	Ellen MacArthur Foundation	Fast-fashion waste measured
2023	PM Vishwakarma scheme includes darzis	India	Support for traditional tailors
2026	Rachel Scott of Diotima named CFDA American Womenswear Designer of the Year	New York	Latest major award (check yearly)

PART 7 — BEYOND GARMENTS

SEW-11 Other sewing

☐ **SEW-11.1 Home furnishing**: curtains, cushion covers, bedsheets, table linen, *gaddi* covers; the easiest straight-line sewing.

☐ **SEW-11.2 Bags**: tote, jhola, pouch, potli, laptop sleeve; boxed corners and straps.

☐ **SEW-11.3 Quilting and patchwork**: layers stitched together. **Godhadi** (Maharashtra and Karnataka quilts of old saris), **kantha** (EMB-07.3), **ralli** (appliqué and patchwork quilts of Sindh, western Rajasthan and Gujarat), **sujani** (EMB-07.4); world: American patchwork, Gee's Bend quilts (Alabama), Japanese **boro**.

☐ **SEW-11.4 Upcycling**: sari into quilt or dupatta, old kurta into cushion, jeans into bag; India's long habit of reuse before the word existed.

☐ **SEW-11.5 Mending and visible mending**: darning, patching, *rafoo* (invisible weaving repair for shawls and silks), sashiko (EMB-08) as decoration; mending shown proudly rather than hidden.

☐ **SEW-11.6 Costume and theatre**: period costume, film costume (MOV), dance costumes for Bharatanatyam and Kathak with pleated fans, cosplay.

☐ **SEW-11.7 Other materials**: leather (*mochi* work), canvas and tents, sails, upholstery, medical stitching (sutures are sewing too).

PART 8 — HISTORY, PEOPLE AND AWARDS

SEW-12 The history of sewing

☐ **SEW-12.1 Prehistory**: awls, then eyed needles at least about 40,000 years ago (SEW-00.01).

☐ **SEW-12.2 Ancient and medieval**: guild tailors in Europe; India's darzi communities serving courts and towns; draped and stitched dress side by side (SEW-00.02).

☐ **SEW-12.3 The machine age**: Thimonnier, Howe, Singer, the "sewing machine war" over patents and the first patent pool (SEW-10).

☐ **SEW-12.4 Ready-made clothing**: uniforms for armies, then shirts and dresses in standard sizes; the sweatshop and the trade union (the ILGWU in New York).

☐ **SEW-12.5 India's tailoring culture**: the neighbourhood darzi, made-to-measure for ordinary people, the blouse-stitching economy around the sari.

☐ **SEW-12.6 India's garment industry**: hubs at Tiruppur (knitwear), Ludhiana, Bengaluru, Delhi NCR, Mumbai, Jaipur and Kolkata; ready-made garment exports of roughly US$ 15 to 16 billion in 2024–25 (AEPC figures; check latest).

☐ **SEW-12.7 Fast and slow fashion**: cheap trend clothes from the 1990s onward; slow fashion, repair and local tailoring as a counter-movement (SEW-00.05).

SEW-13 Couturiers, designers, institutions and awards

☐ **SEW-13.1 World couturiers**: Charles Frederick Worth (haute couture), Coco Chanel, Madeleine Vionnet, Cristóbal Balenciaga, Christian Dior, Yves Saint Laurent, Issey Miyake, Vivienne Westwood.

☐ **SEW-13.2 Indian designers**: Ritu Kumar (from 1969, reviving hand-block prints; Padma Shri 2013), Rohit Bal (1961–2024), Wendell Rodricks (Goa, Padma Shri 2014; 1960–2020), Tarun Tahiliani, Manish Malhotra (film costume), Sabyasachi Mukherjee, Anita Dongre, Rahul Mishra (Woolmark Prize 2014; shows at Paris Haute Couture Week), Gaurav Gupta.

☐ **SEW-13.3 The unnamed masters**: the karigars and darzis whose names rarely appear on labels; credit them where you can.

☐ **SEW-13.4 Institutions**: NIFT (1986), FDCI (1998), Pearl Academy, ITIs (Sewing Technology and Dress Making trades), Apparel Export Promotion Council (AEPC); abroad, Central Saint Martins, Parsons, Bunka Fashion College, Fashion Institute of Technology.

☐ **SEW-13.5 Fashion awards**: CFDA Fashion Awards (USA; 2026 American Womenswear Designer of the Year: Rachel Scott of Diotima); The Fashion Awards by the British Fashion Council (2025 Designer of the Year: Jonathan Anderson, his third win); LVMH Prize for young designers; International Woolmark Prize.

☐ **SEW-13.6 Skills competitions**: WorldSkills includes **Fashion Technology** (pattern making, cutting and sewing under time limits); India selects competitors through **IndiaSkills** (Skill India, NSDC). Latest medal lists are on worldskills.org and skillindia.gov.in.

☐ **SEW-13.7 Museums**: Victoria and Albert Museum (London), The Met Costume Institute (New York), Kyoto Costume Institute, Calico Museum of Textiles (Ahmedabad), National Museum (New Delhi) costume galleries.

PART 9 — IDEAS: PHILOSOPHY, PSYCHOLOGY AND SOCIETY

SEW-14 Identity, sustainability and livelihood

☐ **SEW-14.1 Identity**: clothes say who we are (region, religion, class, profession); tailoring lets the body, not the size chart, decide.

☐ **SEW-14.2 Sustainability and textile waste**: production doubled 2000–2014 and under 1% is recycled into new clothes (SEW-00.05); repair, alteration and buying less are the strongest steps.

☐ **SEW-14.3 The economics of tailoring**: made-to-measure is cheap in India because labour is cheap; that is good for buyers and hard for tailors. A fair price is part of slow fashion.

☐ **SEW-14.4 Livelihood and empowerment**: stitching is among the most common home-based work for women in India; schemes give machines and training, but steady orders and fair rates matter more than the machine.

☐ **SEW-14.5 Psychology**: sewing as focus and flow, patience, problem-solving, and the pride of "I made this".

☐ **SEW-14.6 Philosophy**: Gandhi's khadi and the spinning wheel as self-rule (*swaraj*); the debate between handmade and machine-made, and who controls the tool (SEW-00.04).

PART 10 — THE PRACTICAL GUIDE

SEW-15 Start sewing

☐ **SEW-15.1 First machine**: a sturdy mechanical electric machine with straight, zigzag and a buttonhole is enough; check local service centres before buying; a well-kept old treadle is also good.

☐ **SEW-15.2 First 10 projects**: (1) sew straight lines on paper; (2) hand-sew a button and a hem; (3) cushion cover; (4) tote bag; (5) drawstring pyjama; (6) petticoat; (7) A-line skirt; (8) straight kurta; (9) kids' frock or simple top; (10) one alteration on your own clothes.

☐ **SEW-15.3 Safety**: fingers away from the needle; switch off before changing needle or foot; never sew over pins; keep shears closed and away from children; unplug the iron; take breaks for back and eyes.

☐ **SEW-15.4 Home studio**: good light from the left, a table at elbow height, an ironing board nearby, labelled boxes for notions, a scrap bin for upcycling.

☐ **SEW-15.5 Small tailoring business**: start with alterations and blouses; keep a measurement register and order book; price for time, not just cloth; learn about MSME (Udyam) registration and the PM Vishwakarma scheme; photograph your work for social media.

☐ **SEW-15.6 Learning**: ITI and polytechnic courses, NIFT short courses, Skill India, YouTube teachers, local tailoring classes; practise more than you watch.

PART 11 — REVIEWS AND VOICES

SEW-16 Critics' Corner, the experts said I say, and My Review Desk

☐ **SEW-16.1 Books**: *Reader's Digest Complete Guide to Sewing*; Claire Shaeffer, *Couture Sewing Techniques*; Helen Joseph-Armstrong, *Patternmaking for Fashion Design*; Winifred Aldrich's metric pattern cutting books; Kate Fletcher on sustainable fashion; Dana Thomas, *Fashionopolis* (2019); Indian cutting-and-tailoring guides in Hindi and regional languages.

☐ **SEW-16.2 Review template**: the machine, book, course or garment in two lines (dated price); what experts said (3 to 5 named views, in my words); what I say (stitch quality, inside finish, fit, ease of use, after-sales service); who made it and were they paid fairly; my rating on a clear five-point scale; tags and try-next links.

PART 12 — NAVIGATION, SOURCES AND NOTES

SEW-17 Tag dictionary and site map

☐ **SEW-17.1 Tags**: #sewing; machine tags (#handcrank #treadle #electric #computerised #overlocker #industrial); technique tags (#seams #darts #hems #zips #collars #sleeves #pressing); garment tags (#kurta #salwar #churidar #blouse #petticoat #lehenga #sherwani #shirt #trousers #jeans #dress); other tags (#quilting #godhadi #ralli #upcycling #mending #costume); theme tags (#history #fastfashion #slowfashion #livelihood #women #tailor).

☐ **SEW-17.2 Cross-links**: Crochet (CRO), Knitting (KNT), Embroidery (EMB); Technology (textile machines: TEC); Movies (costume design: MOV); the Curriculum Index (Home Science, Economics, History) and the Exam Guide (questions on the textile industry, khadi and Skill India).

☐ **SEW-17.3 Web addresses**: /sewing/ (home, Gateway), /sewing/needle-with-a-hole/<leaf>/ , /sewing/machines/<type>/ , /sewing/tools/<tool>/ , /sewing/fabrics/<fabric>/ , /sewing/techniques/<technique>/ , /sewing/patterns/<topic>/ , /sewing/garments/<garment>/ , /sewing/quilting/<style>/ , /sewing/history/<era>/ , /sewing/people/<name>/ , /sewing/awards/<award>/<year>/ , /sewing/ideas/<topic>/ , /sewing/guide/<topic>/ , /sewing/reviews/<item>/ .

☐ **SEW-17.4 Search phrases to own**: "how to stitch a kurta at home", "blouse cutting and stitching", "sewing machine for beginners India", "types of seams", "sewing machine thread bunching fix", "history of sewing machine", "what is slow fashion", with Hindi lines ("kurta kaise silen", "blouse ki cutting", "silai machine kaise chalayen", "silai sikhna shuru kaise karen").

SEW-S Sources and credits

☐ **India**: Ministry of Textiles; NIFT; Apparel Export Promotion Council; Bureau of Indian Standards (household sewing machine standards); Khadi and Village Industries Commission; Skill India and NSDC; PM Vishwakarma portal; Usha International and Singer India histories.

☐ **World**: Smithsonian (National Museum of American History sewing machine collection); Science Museum London; Victoria and Albert Museum; The Met Costume Institute; Ellen MacArthur Foundation, *A New Textiles Economy* (2017); Fashion Revolution; WorldSkills International; CFDA; British Fashion Council.

☐ **Research**: Gilligan and colleagues, "Paleolithic eyed needles and the evolution of dress", *Science Advances* (2024); archaeological reports on Denisova Cave.

SEW-E Editor's notes

1. **Oldest needle**: written as "at least about 40,000 years", with the Denisova date of about 50,000 years marked as debated.
2. **Blouse history**: the Jnanadanandini Devi story is given as the common account, not settled fact.
3. **Gandhi on Singer**: marked as reported, since the exact wording varies between sources.
4. **Export figures** and **award winners** are dated (2024–25, 2025, 2026); check each year.
5. **Times in SEW-09** are rough guides for a confident sewer; beginners should double them.
6. **Added beyond the brief**: Weisenthal 1755, the patent pool, Butterick, Triangle fire, rafoo, Gee's Bend, boro, PM Vishwakarma, the darzi's alteration allowance.

SEW-T My Sewing Tracker

Tick ☐ as you learn each technique and make each garment. Keep a sewing log: date, project, fabric, pattern or measurements, what went wrong, what I would change, photo.

Progress tracker

Part	Sections	Explored ☐	Started	Finished
PART 0 — THE INTRODUCTION TWIG:	7			

HOW DID A NEEDLE WITH A HOLE IN IT CHANGE HUMAN HISTORY?				
PART 1 — THE MASTER MAP OF SEWING	1			
PART 2 — SEWING MACHINES	2			
PART 3 — TOOLS AND MATERIALS	2			
PART 4 — TECHNIQUES	2			
PART 5 — PATTERN MAKING AND FIT	1			
PART 6 — REGISTERS	2			
PART 7 — BEYOND GARMENTS	1			
PART 8 — HISTORY, PEOPLE AND AWARDS	2			
PART 9 — IDEAS: PHILOSOPHY, PSYCHOLOGY AND SOCIETY	1			
PART 10 — THE PRACTICAL GUIDE	1			
PART 11 — REVIEWS AND VOICES	1			
PART 12 — NAVIGATION, SOURCES AND NOTES	4			